Tidal gravity across interrupted optical clock comparisons

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Optical clocks compare gravitational potential through its effect on the passage of time.[1-3] Extending this measurement to time-dependent signals requires separating gravitational responses from variations of the clocks and their frequency-transfer systems. Here we analyse independently acquired comparisons of a strontium clock in Shanghai and a ytterbium clock in Wuhan, connected by a 1,350-km fibre route.[4] Eight observation periods spanning 353 days yield 3,266 non-overlapping 600-s averages. A common scale multiplying the predicted tidal signal is 0.95 ± 0.11 with fixed ten-day calendar baselines. Allowing independent 24-h and 12-h variations leaves an M2+N2 lunar semidiurnal scale of 0.82 ± 0.29, with errors summarized over specified statistical working models. Removing the fitted whole template reduces the M2-frequency harmonic power by 98%, and a held-out-period diagnostic improves in all eight periods. A matched round-trip fibre record quantifies one monitored transfer contribution. Tests retaining the measured frequency levels do not identify an attributable seasonal solar-tide signal. Together with contemporaneous independent clock-network measurements,[5] these observations support dynamic gravitational-potential sensing and distinguish the calibration requirements of semidiurnal and seasonal measurements.

The gravitational redshift makes a clock's rate a local probe of gravitational potential. Optical-clock performance and coherent frequency transfer have enabled remote potential comparisons and motivated their use in geodesy.[1–3] The potentials of the Moon and Sun, together with the Earth's deformation and mass redistribution, add a time-dependent contribution. A distant-clock comparison therefore contains a predicted astronomical signal as well as the clocks' intrinsic frequency ratio, instrumental noise and systematic shifts. Measuring this signal tests the response to changing potential rather than only the mean difference between two sites.

For an intermittently operated comparison, elapsed calendar time and sensitivity to a tidal component are different quantities. A signal can be constrained through its variation within observed intervals even when the frequency levels cannot be transferred accurately across all interruptions. Conversely, a long calendar span need not constrain a slow signal if clock corrections or instrumental changes are uncertain. Daily environmental variations create a further distinction: a response to the full predicted tide and a response to lunar semidiurnal components after allowing arbitrary solar-day terms answer different questions.

The contemporaneous European measurements of Stahl and colleagues track tidal potential variations with a clock network and Kalman estimation.[5] Our independently acquired Shanghai–Wuhan comparisons address the response across widely separated observation periods. Our earlier ratio preprint reported a tidal correlation in the summer record.[4] Here we analyse the broader interrupted record, establish a whole-template benchmark and test how lunar and longer-period responses depend on solar-day backgrounds and preservation of the measured frequency levels. Table 1 distinguishes the overlapping observations and the statistical targets of the earlier and present analyses.

A common response across observation periods

The reconstructed record contains 1,952,992 source measurements in 3,266 non-overlapping averages, drawn from 54 effective intervals and eight observation periods between 7 September 2025 and 26 August 2026. The represented measurements occur on 39 BJT calendar days. Frequency conversion follows the recorded year-specific comb and electronic configurations, and clock corrections follow the corrected experimental ledger (Methods). The resulting ratios preserve the relative levels of the measured intervals. They are expressed relative to one numerical reference ratio; this coordinate choice is not a new absolute-frequency-ratio determination.

We first fit a single signed scale, A_T, multiplying the entire predicted fractional-frequency tide $T(t)$,

$$y(t) = A_T T(t) + \gamma_{b(t)} + \epsilon(t),$$

where $b(t)$ assigns each observation to a fixed ten-day BJT calendar block. Thirteen such blocks contain observations. The common amplitude is estimated jointly with these baselines; the measured points are not shifted individually or aligned to the tidal prediction. The baselines are intended for the short-period comparison and are not used in the seasonal analysis below.

The result is $A_T = 0.9467$, with residual-scaled least-squares, delete-day and delete-period standard errors of 0.0869, 0.0964 and 0.1076, respectively. We summarize this result as 0.95 ± 0.11, retaining the largest of these standard-error estimates. Five-day calendar blocks give 0.9488 ± 0.1076. These nearby estimates establish stability to this specified change of baseline duration; they are correlated estimates from the same observations. An exploratory stress test adds a separate linear drift within every occupied calendar block, without changing the observations. The ten-day whole-template scale becomes 0.902 ± 0.090 and the five-day scale 0.893 ± 0.099; the ten-day change is -0.045 ± 0.048 using paired day deletions. Thus the common response is also retained under this specified allowance for slow within-block drift.

Figure 1a shows a 48-h window selected by the largest number of retained source members, independently of its agreement with the tidal model. Every 600-s observation is shown, together with a three-hour display average. The template amplitude in this panel is estimated from the other seven observation periods. Only the held-out period's baseline is estimated within that period. Its measured points retain their frequency levels relative to the same global reference used for the full record.

The frequency-domain diagnostic uses the actual observation times and the same nuisance projection before and after template subtraction. At the M2 frequency, the harmonic variance-reduction statistic decreases from 14.87×10^{-36} to 0.31×10^{-36}, a reduction of 97.9% (Fig. 1b). This is a phase-sensitive subtraction diagnostic, because the model is removed in the time domain before the harmonic scan. The scan itself is descriptive and is not assigned an additional discovery probability.

We also omit each observation period in turn, estimate A_T from the remaining periods and apply that amplitude to the held-out data while fitting their prescribed nuisance baselines. The training amplitudes range from 0.883 to 1.029. Residual sums of squares decrease in all eight held-out periods, by 3.42% when pooled (Fig. 1c). This tests transfer of the amplitude between periods, conditional on their fitted baselines; absolute interperiod calibration is examined separately below.

Separating lunar tides from solar-day variation

The whole-template test uses the predicted relative amplitudes and phases of all tidal components. We next allow sine and cosine terms at exactly 24 h and 12 h independently within each ten-day block. These terms represent possible solar-day variations without identifying their physical source. The full-template scale becomes 0.74 ± 0.28; the M2-frequency diagnostic still decreases by 93.1% after subtraction. With five-day blocks, the scale is 0.83 ± 0.37. The reduced precision reflects the overlap between tidal and nuisance variations on the observed support.

To isolate the lunar semidiurnal response, we fit a common scale for M2+N2 while giving the near-12-h and diurnal tidal groups independent scales. The same free solar-day terms are retained. The ten-day result is $A_L = 0.8216$, with standard errors 0.2493, 0.2851 and 0.2269 from least squares, day deletion and period deletion. The corresponding five-day estimate is 0.80 ± 0.42. Thus the response remains compatible with the expected scale when the exact solar-day terms are fitted freely (Fig. 2a). Adding within-block linear drifts gives 0.83 ± 0.30 for the ten-day lunar design and 0.73 ± 0.56 for five days, retaining the central response while exposing the lower precision of the shorter-block design.

Allowing a common lunar quadrature tests consistency with the predicted phase without shifting measurement times. The ten-day in-phase and quadrature coefficients are $(0.792, 0.129)$, with resultant scale 0.802. The unit, in-phase prediction lies within the nominal delete-day confidence regions for both five-day and ten-day baselines (Fig. 2b). A direct-potential sensitivity calculation, retaining the ocean-loading prediction and the same observation members, changes the lunar scale by less than 0.005 and the whole-template scale by less than 0.010 across the three specified alternative solid-tide approximations (Methods). Fitting the two constituents separately gives M2 0.88 ± 0.31 and N2 0.16 ± 1.09 in the ten-day design. N2 is not independently resolved by this record.

The information supporting these estimates is uneven. Summer contributes 74.7% of the squared lunar template after projection against the other fitted columns, compared with 38.7% for the whole-template model without free daily terms. These are properties of the sampled least-squares designs, not physical fractions of the tidal signal or eight independent seasonal replications. Free daily terms also weaken the held-out diagnostic: four of eight periods improve, and the pooled reduction is 0.323% (Fig. 1c).

The uncertainty depends on the permitted correlation structure. A restricted wild bootstrap over the eight periods gives two-sided zero-response probabilities of 0.0078 and 0.0313 for the ten-day whole-template fits without and with free daily terms. The corresponding five-day fit with daily terms gives

0.1016. For M2+N2, the ten-day period-wise test gives 0.1484, whereas a fitted Gaussian white-plus-Ornstein–Uhlenbeck model gives 0.00098. These tests make different assumptions and have different power. We retain that distinction instead of assigning a single detection significance to the lunar estimate. Simulations that repeat raw-time screening and averaging demonstrate why correlated-noise checks matter: under the specified 12-h correlated-noise scenario, least-squares 95% coverage is 73–77%, whereas delete-day coverage is 95–98% (Methods).

The measured transfer control

The recorded round-trip monitor provides a simultaneous test of one part of the frequency-transfer system. We designate channel B2 from the apparatus record and pair it with the clock measurements without interpolation. A 600-s bin enters the regression only if all its original source members have unique valid monitor observations. The complete monitor record contains gross frequency excursions, so we retain the unfiltered result and compare fixed windows of ±10, ±100 and ±1000 Hz around the recorded 10-MHz monitor carrier. This quality check changes the monitor-comparison support, not the main clock result.

At zero nominal time shift, the ±100-Hz criterion leaves 1,519 complete bins with 910,041 source members. On all simultaneous valid second samples, the clock and nominal-optical-normalized round-trip monitor have one-hour overlapping Allan deviations of 7.66×10^{-18} and 6.26×10^{-18}, respectively (Fig. 3a). These are measured channel diagnostics; no assumed factor of two converts the monitor into a calibrated one-way transfer bound.

On the matched bin support, adding B2 as a regressor changes the whole-template clock scale by 0.0057 ± 0.0073 without free daily terms and 0.0415 ± 0.0345 with them, where the errors are paired delete-day standard errors. Figure 3b shows the corresponding nominal 95% intervals and threshold sensitivity. The complete analysis also retains fixed ±5-s pairing alternatives. This comparison quantifies the tested linear monitor association. It does not observe every laboratory clock-to-link branch, and it does not exclude clock effects or unmonitored transfer contributions.

Longer periods require retained frequency levels

For Mf, Mm, Msf and the annual and semiannual seasonal constituents Sa and Ssa, we retain the measured levels and fit only one intercept across the entire record (Fig. 4a). The five long-period components are fitted jointly as free quadratures, together with short-period tides and global 24-h and 12-h terms. A single global linear drift is examined separately. No short-period block baselines are transferred into these seasonal tests.

The no-drift model does not identify an attributable seasonal solar-tide signal. Its nominal delete-period 95% upper bounds on free-phase fractional-frequency amplitude are 1.30×10^{-16} for Mf, 4.44×10^{-16} for Mm, 2.48×10^{-16} for Msf, 5.53×10^{-16} for Sa and 3.60×10^{-16} for Ssa (Fig. 4b). They are conditional statistical bounds, rather than unconditional bounds on gravitational response: incomplete clock corrections and unbounded residual calibration changes remain relevant. Adding a global drift produces strongly model-dependent empirical seasonal terms whose magnitudes greatly exceed the predicted gravitational amplitudes.

An independent degree-two calculation of the diurnally averaged solar potential gives a semiannual differential response of approximately 2.02×10^{-20} for the two sites. Their similar latitudes suppress this component. The observed long-period sensitivity is therefore far from resolving the predicted seasonal signal. In addition, the current ocean-loading synthesis does not contain the five long-period constituents tested here. Preserving levels makes these limitations visible; extending the calendar span alone does not remove them.

Discussion

The Shanghai–Wuhan measurements recover a common response to the predicted tide across separated observation periods. The whole-template amplitude, reduction of the semidiurnal spectral feature and held-out-period diagnostic support this interpretation within the stated baseline and noise models. Explicit solar-day terms retain a lunar semidiurnal response compatible with the model, with substantially less information and more dependence on the statistical assumptions. The measured monitor constrains a specified transfer association, while leaving unmonitored paths and clock variations within the interpretation boundary.

The earlier ratio result and the present analysis are complementary uses of overlapping observations. Table 1 separates their data, estimands and errors. The earlier $0.87(15)$ whole-template result and its reported significance cannot be transferred directly to the present lunar-component estimate. Conversely, the more general nuisance and noise tests here do not by themselves invalidate that earlier, narrower statistical statement. In particular, the historical 124-of-200 injection check concerns a one-standard-error interval, not a 95% interval; its finite-simulation uncertainty includes the nominal Gaussian one-standard-deviation coverage.

The contemporaneous European work establishes tidal tracking with a network, Kalman estimator and same-site controls.[5] Our independent observations test a common response across interrupted operation and quantify what remains when solar-day variations are fitted freely. Preserving the measured levels then exposes the separate requirements for longer-period sensing. These results connect a common physical benchmark to complementary measurement designs. Both experiments test clock-based sensing of time-dependent potential against predicted tidal responses; the present combined-template fit and harmonic decomposition do not separately estimate geophysical response parameters.

The practical implication is that dynamic chronometric comparisons have more than one calibration timescale. Semidiurnal variations can be tested within observation blocks while their common response is assessed across periods. Seasonal inference requires the relative levels themselves to remain calibrated. Improving clock and local-transfer monitoring therefore strengthens both the attribution of the short-period response and the prospect of measuring slower potential changes.

References

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Table 1

Table 1 | Related tidal estimates and their statistical targets. The estimates share observations and are not combined. Values in parentheses or after ± denote the error scale specified in the last column; they are not a common calibrated confidence level.

Analysis	Observations and averaging	Tidal target and nuisance model	Scale	Error definition
Hou et al.[4]	Summer record; 16 intervals; non-overlapping 1,200-s inference windows	Whole predicted tide; interval-centred fits and precision weighting	0.87(15)	Reported weighted statistical error; reported 5.7σ under that analysis
Previous component analysis, v10	3,186 overlapping filtered outputs; 54 interval offsets	M2+N2; campaign-specific 24-h/12-h terms	0.86 ± 0.18	max(LS, delete-day SE)
Same v10 support, whole-template comparison	Identical 3,186 outputs, offsets and daily terms	Whole predicted tide; same offsets and daily terms	0.89 ± 0.20	max(LS, delete-day, delete-period SE)
Present reconstruction	3,266 non-overlapping 600-s means	Whole predicted tide; ten-day calendar baselines	0.95 ± 0.11	max(LS, delete-day, delete-period SE)
Present daily-term control	Same 3,266 means	Whole predicted tide; ten-day baselines and 24-h/12-h terms	0.74 ± 0.28	Same standard-error summary
Present lunar decomposition	Same 3,266 means	M2+N2; independent other tide groups and ten-day daily terms	0.82 ± 0.29	Same standard-error summary

Figure legends

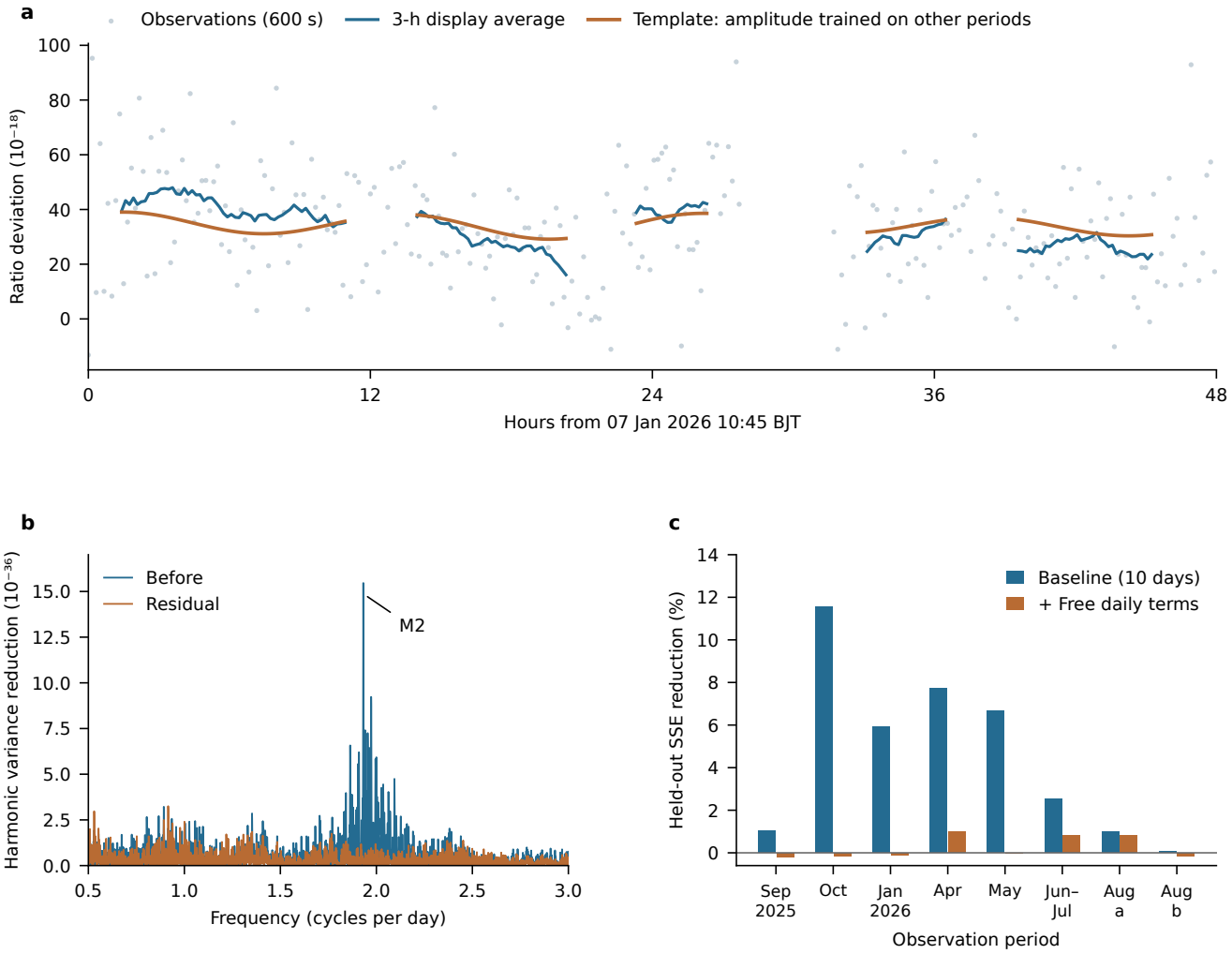


Fig. 1 | A common tidal-template response across interrupted observations. a, The 48-h window containing the largest number of retained source members (158,072 in 264 bins), selected without reference to fit quality. Points are 600-s averages after subtraction of one common reference used for the whole record. The blue curve averages 18 consecutive bins within uninterrupted supports for display. The orange curve applies the same display average to the template, with its amplitude trained on the other seven periods and its baseline fitted in the held-out period. No individual segment offsets are subtracted from the points. **b,** Actual-time harmonic scans of all 3,266 bins before and after whole-template subtraction, with the same ten-day baseline projection. The ordinate is harmonic variance reduction per observation, not a physical power spectral density or a significance threshold. **c,** Changes in held-out-period residual sums of squares when the amplitude is learned from the other periods. Blue and orange denote the baseline-only and free-daily-term designs. The held-out nuisance terms are fitted locally; the eight folds share training data. All time-series points are repeated observations of the same apparatus, not independent experiments.

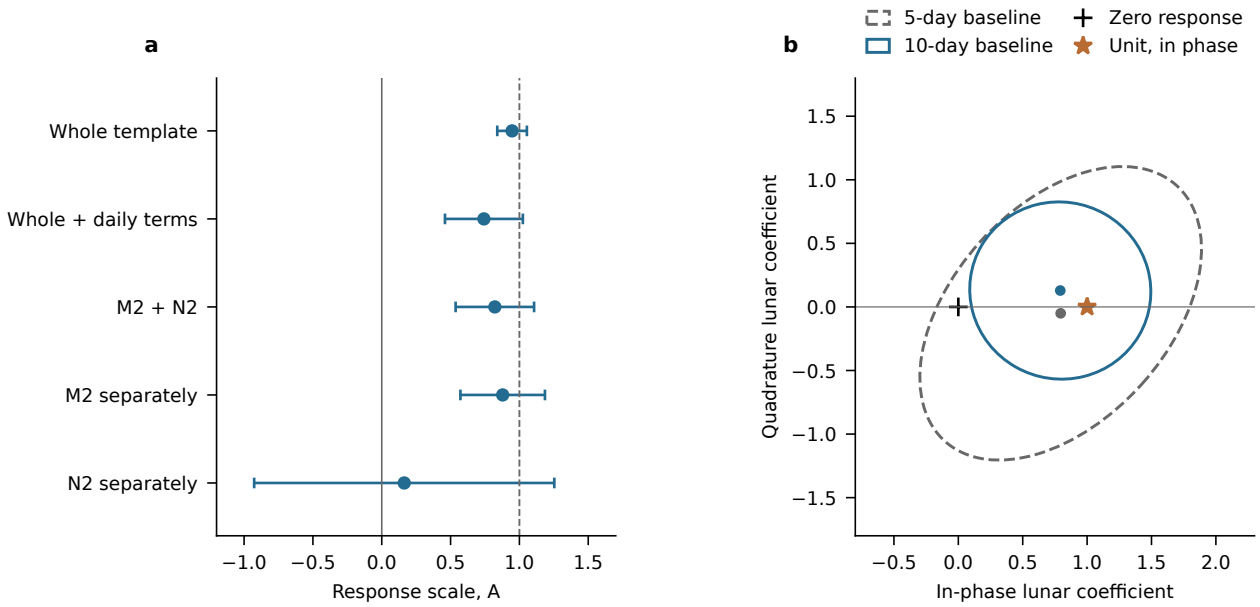


Fig. 2 | Lunar response after allowing solar-day variation. **a,** Scales estimated from the same 3,266 averages. All rows except the first include ten-day 24-h and 12-h nuisance terms. M2 and N2 are estimated jointly with independent scales in the last two rows. Bars show one standard-error summary, max(LS, delete-day, delete-period); the vertical references mark zero and unit response. The rows are correlated estimates of different targets. **b,** In-phase and quadrature coefficients of a common lunar response for five-day and ten-day baselines. Curves are nominal 95% delete-day working-model regions for the two coefficients; points mark their estimates. The cross and star denote zero and the unit, in-phase prediction. No observation times or phases are adjusted. These individual regions do not include a correction for the exploratory model comparisons, and their nominal coverage is not assumed under arbitrary noise.

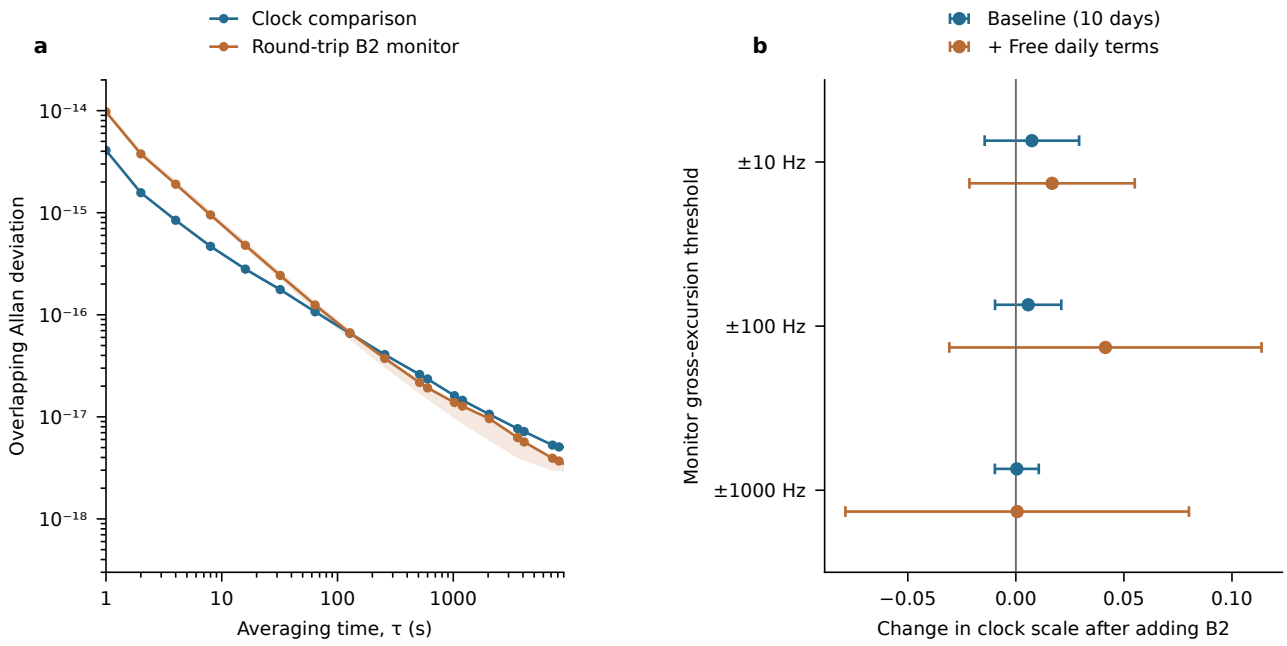


Fig. 3 | Scope of the measured round-trip transfer control. **a,** Overlapping Allan deviations of the clock comparison and predesignated B2 monitor on simultaneous valid second samples. Solid curves use the ±100-Hz gross-excursion criterion (913,692 common seconds before averaging-time run requirements). Shading spans the ±10-Hz and ±1000-Hz sensitivity results as well as the main curve; it is not a confidence band. No differences cross gaps, and contributing runs must span at least four averaging times. The monitor is normalized by the nominal optical carrier without an assumed one-way conversion. **b,** Paired changes in the clock's whole-template scale on adding B2, at zero nominal pairing shift. Thresholds leave 1,476, 1,519 and 1,520 complete 600-s bins. Error bars are nominal 95% delete-day intervals for the paired changes, with 19 degrees of freedom; they are not interval differences computed from independent fits. Baseline-only and free-daily-term designs use the same support within each threshold. The control covers the recorded channel and time overlap, including neither unmonitored laboratory branches nor the complete year.

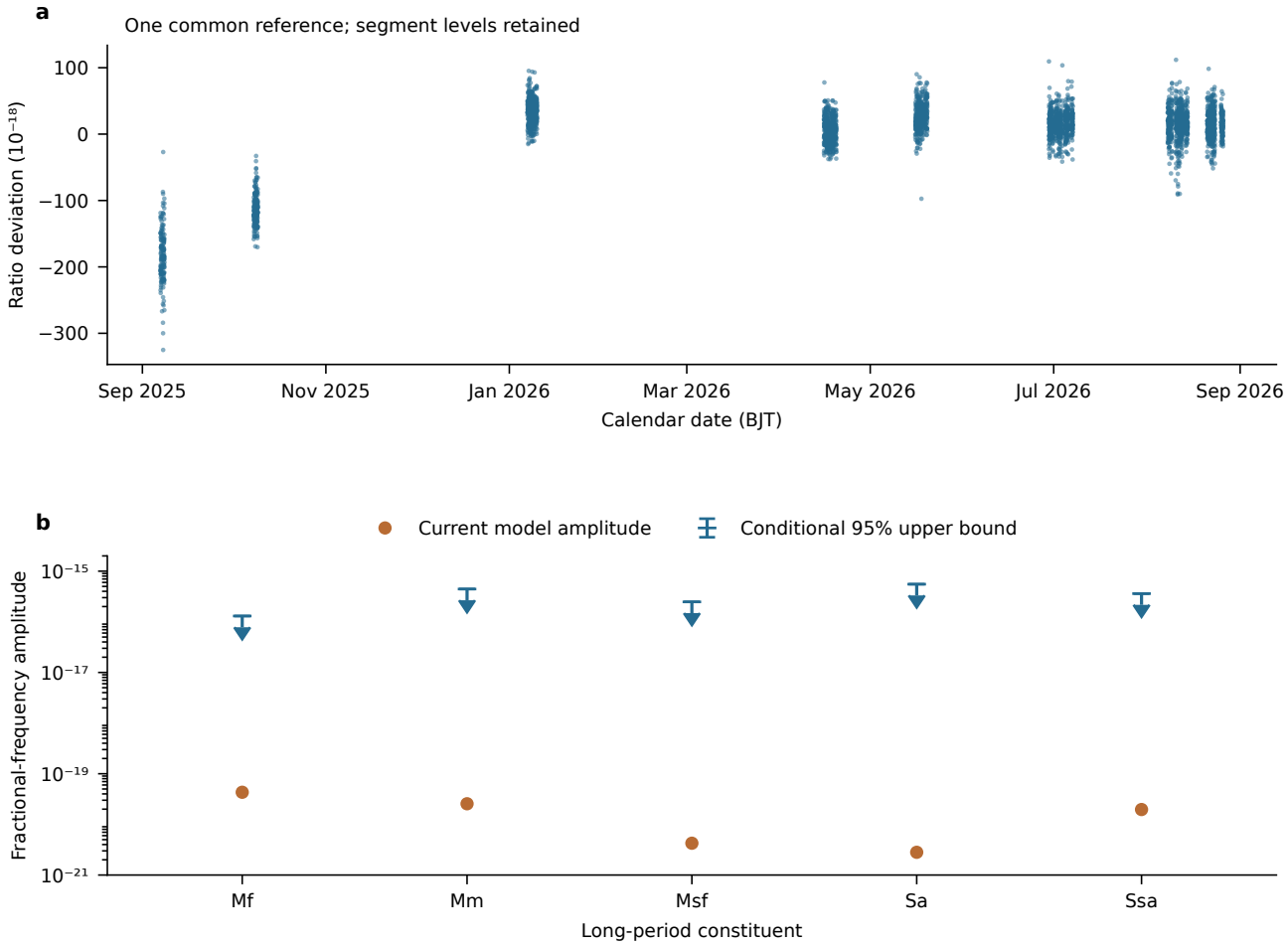


Fig. 4 | Preserved frequency levels and long-period sensitivity. a, All 3,266 reconstructed ratio averages, with one common plotting reference subtracted. No fitted segment or period centres are removed. Points retain the actual gaps and period-to-period level differences. **b,** Nominal 95% upper bounds on free-phase amplitudes from the full-record, single-intercept model with joint short-period and global daily terms, without a global drift. Caps indicate upper bounds and arrows indicate the permitted lower-amplitude direction; arrow lengths are graphical. Bounds use the largest eigenvalue of each two-coefficient delete-period covariance and an approximate two-dimensional region with seven denominator degrees of freedom. They are individual bounds, conditional on the stated calibration and noise assumptions. Orange points give amplitudes from the current theoretical decomposition, not measured signals. The five-constituent family bounds and drift sensitivity are specified in Methods. Unknown calibration terms and incomplete long-period loading prevent these bounds from being interpreted as unconditional gravitational limits.

Methods

Measurement coordinate and frequency conversion

We use the year-specific conversion recorded for the Shanghai Sr and Wuhan Yb comparison.[4] For 2025, with $N_{1397} = 858456$, $N_{1550} = 773598$, $N_{1156} = 1295739$ and $N_{1550,\mathrm{WH}} = 966996$,

$$f_{1397} = \frac{f_{\mathrm{Sr}}}{2(1-\alpha)}, \quad \alpha = \frac{10.93629485}{25N_{1397}}, \quad f_{1550,\mathrm{WH}} = \left(N_{1550} + \frac{8}{25}\right)\frac{f_{1397}}{N_{1397}}.$$

The eight Shanghai reference units include the seven in the transmitted carrier and the additional unit at Wuhan. The Wuhan repetition frequency is $f_{\mathrm{Yb}}/(2N_{1156})$, and its 10-MHz reference is $f_{\mathrm{Yb}}/(40N_{1156})$. The comb offset is 26 such reference units. If the recorded counter value q is in nominal hertz realized by this Wuhan reference, the locked ratio is

$$R_{\mathrm{lock}} = \frac{N_{1156}}{N_{1397}}\frac{N_{1550} + 8/25}{N_{1550,\mathrm{WH}} + 26/20 + dq/(20\times 10^7)}\frac{1}{1-\alpha}.$$

The reconstruction uses $d = 2$ and the stated α in 2025, and the separately recorded 2026 convention $d = 1$, $\alpha = 0$. A 2025 reading near 66 MHz represents a divided beat near 132 MHz; the recorded 2026 configuration is near 33.6 MHz. The counter-reference convention is an explicit experimental assumption, not a consequence of a reference being labelled 10 MHz. The electronic divider is restored once and is unrelated to a round-trip fibre correction.

With signed clock shifts a for Sr and b for Yb, we form $R = R_{\mathrm{lock}}(1+a)/(1+b)\,(1+\Delta G)$, with the inherited constant $\Delta G = -3.116\times 10^{-15}$. The coordinate is $y = (R/R_{\mathrm{ref}} - 1)10^{18}$, with $R_{\mathrm{ref}} = 1.2075070393433377166$. Decimal anchors prevent loss of small ratio changes through subtraction in double precision. The constant potential correction and ratio anchor do not constitute a new static-geopotential or absolute-ratio measurement.

The eight periods are September and October 2025; January, April and May 2026; June–July 2026; and early and late August 2026. In units of 10^{-18}, the corrected Sr density entries are −0.43, −0.41, −0.30, −0.33, −0.25, −0.24, −0.33 and −0.29. In the same units, the first six Zeeman corrections are −178.50, −178.87, −179.14, −180.19, −180.18 and −178.57; August Zeeman and all blackbody corrections are recorded as online. The background-gas entry is -0.67×10^{-18}. AC entries for the first five periods are missing; the final three are $[7.44, 8.93, 9.03]\times 10^{-18}$. The documented positive 37.2-mHz feedback offset is applied to the first three periods at the assumed 698-nm optical-frequency node, giving 86.6119×10^{-18} using the nominal wavelength. The node interpretation is retained as a condition. Printed Yb shifts for original groups 1–9 are used for the principal reconstruction, with the component sums examined separately. The May Yb correction is missing. Unknown entries represented as zero in the computational coordinate are not measured zero shifts.

Observation selection and averaging

The corrected experimental intervals select 1,966,912 unique source rows from 42 files, retaining fractional-second BJT labels. Intervals include the first displayed PDF second and exclude the second immediately after the last displayed second. Rows are sorted by time, file and source row and deduplicated by source identity. Finite beat offsets below 1 MHz in magnitude and strictly within 10 Hz of the plausible-sample median leave 1,966,908 rows. The three-row difference from the experimental table's count is retained with boundary provenance.

Within each effective interval, 300-s block means are compared with a 13-block running median using reflected boundaries. The median absolute deviation of the block residuals, divided by 0.6744897501960817, defines a screening scale. Blocks exceeding six scales in absolute residual and their immediate neighbours are excluded. This rule uses the measured beat, not a fitted tide residual. It removes 11,131 additional rows. Non-overlapping 600-s means require at least 300 retained members; 3,266 means contain 1,952,992 members. All tidal and nuisance columns are averaged over these exact members. Observations are not interpolated. Day deletions use bin-mean BJT days; 16 bins cross midnight, so this is not deletion of every raw sample from the named civil day.

Tidal prediction and component design

The common prediction combines solid-Earth and ocean-loading series on a 90-s grid. The solid-Earth contribution uses pysolid displacements under the recorded IERS 2010 implementation,[6] converted as

$$s(t) = -\frac{1 + k_2 - h_2}{h_2}\frac{g[u_{\mathrm{WUH}}(t) - u_{\mathrm{SHA}}(t)]}{c^2},$$

with $h_2 = 0.6078$, $k_2 = 0.3010$ and $g = 9.794\,\mathrm{m\,s^{-2}}$. This is a displacement-based approximation, not a complete independent degree-dependent potential calculation. Ocean loading uses 18 constituents from grids locally documented as FES2022b,[7] convolved with PREM load-Love-number Green functions containing $1+k_n'-h_n'$. The archived Han–Wahr table[8] through degree 696 is extended to degree 40000, with a cosine taper over the final 30% and the recorded centre-of-solid-Earth degree-one convention. It includes direct water potential, induced potential and displacement.

For decomposition, 360,001 theory samples are projected onto quadratures of Q1, O1, P1, K1, N2, M2, S2, K2, Mf, Mm, Msf, Sa and Ssa plus a constant. The rank is 27. The unfitted remainder is retained. BJT is converted to UTC once; the astronomical implementation uses the saved FES convention and 69.184-s TT–UTC offset. Full-template fitting multiplies the complete original theory and does not subtract any of it beforehand. Component fitting retains the original relative phases within the grouped template and subtracts the remaining fixed prediction once.

In fractional-frequency units of 10^{-18}, the lunar design is

$$y - F = A_L C_L + a_{12} C_{12} + a_D C_D + Z_b \gamma + Z_{24,12}\beta + \epsilon.$$

C_L contains M2+N2, C_{12} contains S2+K2, and C_D contains the four diurnal constituents. F contains the rest of the theory, including the long-period terms. The daily design has sine and cosine columns at one and two cycles per day in each occupied block. The ten-day lunar model has 68 parameters and full rank. The blocks are fixed by $\lfloor t/(10 \times 86400) \rfloor$ on the BJT-labelled epoch coordinate; five-day sensitivity uses the analogous rule. These are calendar blocks, irrespective of actual uptime. The S2+K2 scale is not interpreted as separately resolved from the exact 12-h nuisance terms. The exploratory slope test adds one linear column in each occupied block, centred on that block's observed mean time and divided by its fixed duration. All six constant-versus-slope comparisons retain identical data and templates. Paired deletion differences estimate the uncertainty of each change in scale. These short-period baselines and slopes are not used for long-period inference.

Three theory sensitivities replace only the solid-Earth contribution with direct geometric Sun–Moon potentials from DE430: degree two with constant response $1 + k_2 - h_2$; degrees two and three with their respective constant responses; and degrees two and three with nominal order-dependent real k_{nm} from IERS Table 6.3.[6] The third-degree displacement response is $h_3 = 0.292$. Fixed station coordinates are transformed using IAU 2006/2000A rotation, with UT1 approximated by UTC and zero polar motion. These controlled approximations omit full frequency-dependent and complex responses and are not replacements for a complete IERS potential implementation. On the identical 3,266 member-matched observations, the three alternatives give whole-template scales 0.942, 0.937 and 0.937, and lunar scales 0.826, 0.817 and 0.817. All alternative results are retained; the primary template is unchanged.

Estimation, correlation and validation

Least squares uses column-normalized singular-value decomposition. Numerical scaling does not change the objective. We retain residual-scaled LS covariance, delete-day and delete-period jackknife covariance, and elapsed-time Bartlett HAC sensitivities with one-, three- and ten-day bandwidths. For G deletion groups, $C_{\mathrm{JK}} = (G-1)G^{-1}\sum_g(\hat{\theta}_{-g} - \bar{\theta})(\hat{\theta}_{-g} - \bar{\theta})^T$. Unsupported nuisance columns are removed in a deletion; absent target columns or rank failures are recorded. The main error summary is the largest LS, day and period standard error. A componentwise maximum is not itself a joint covariance or a calibrated confidence region.

The restricted wild bootstrap fits the tested null, multiplies residuals by a common ±1 sign within each of eight periods, refits and studentizes with the delete-period standard error.[9] All 256 sign patterns are enumerated. Period independence and symmetric errors remain assumptions; estimating nuisance residuals prevents this from being an exact randomization test. The 0.0078125 minimum two-sided

outcome reflects the finite enumeration. The tests are exploratory and do not correct for every analysis choice.

For lunar noise sensitivity, a finite covariance grid combines white noise with an OU process over actual elapsed times, using correlation fractions 0.1–0.995 and time constants 10–5120 min, plus the white-only case. Null and alternative fits each reselect the grid point. The ten-day fit selects white noise by maximum likelihood, whereas restricted likelihood selects a weak, long-correlated boundary solution. The 4,096 null simulations reselect the same grid; three exceed the observed likelihood-ratio statistic, giving $(3 + 1)/(4096 + 1) = 0.000976$. This probability is conditional on the fitted Gaussian family. The period-wise test's weaker evidence is retained alongside it; simulations show low power when information is concentrated in a few periods.

Fixed-support simulations use 1,024 realizations per specified noise scenario. A further 2,048 realizations repeat raw-time validity and transient screening, member-matched averaging and the three ten-day fits. Four scenarios, 512 each, use white frequency noise of per-second standard deviation 550; added white-phase differences with endpoint scale 2300; added OU noise of 12-h scale and stationary standard deviation 15; or added randomly signed 900-s transients of amplitude 1500 at an average of one per observation day, in 10^{-18} frequency units where applicable. The injected whole-template and tidal-group scales are one. The experimental intervals and calibration conventions remain fixed. Seed 20261105 and separate scenario/trial streams are recorded. No numerical failures occur in the 6,144 fits. These tests examine specified noise and processing assumptions, not arbitrary systematic errors or extreme null tails.

The earlier component reduction used overlapping 1,200-s triangular outputs and 54 interval offsets. Its saved injection diagnostic counted $|\widehat{A} - 1| \le u$ in 124 of 200 realizations. Its one-standard-error coverage estimate of 0.62 has a Wilson 95% interval [0.5511, 0.6844], including approximately 0.6827. The bias was +0.0275 and the coefficient SD was 0.1844. This fixed-support test did not repeat the original selection or certify the earlier preprint's reported extreme-tail significance.

Spectral and held-out diagnostics

For a nuisance matrix Z, both the observed response and each trial harmonic are projected with $M_Z = I - ZZ^+$. At each frequency, the diagnostic is the reduction in residual sum of squares obtained from its two quadratures, divided by the number of observations. The scan uses actual bin-mean times from 0.25 to 4 cycles per day, at a spacing of $1/(5\Delta t)$, and includes the specified tidal frequencies. It is repeated after subtracting $\widehat{A}_T M_Z T$, keeping the same projection. The sampling window is computed from the same times. Gaps are never concatenated into continuous data. The scan is not a physical PSD estimate or a calibrated multiple-frequency significance test.

For leave-period-out diagnostics, the amplitude is learned from the seven training periods. Within the test period, both its response and template are projected against its own prescribed nuisance columns. We compare their sums of squares before and after applying the training amplitude. This fixes the test amplitude but estimates test nuisance terms; the folds share observations and are not independent replications.

Fibre monitor and stability

The 29 recorded monitor files are interpreted according to the experimental B2 designation. Their BJT-labelled nominal one-second axes use the floored first file timestamp plus row index, matched to the corresponding nominal clock axes; repeated or nonfinite monitor seconds are excluded. Fixed pairings at −5, 0 and +5 s test sensitivity to that convention, without certifying hardware synchronization or logging latency. Complete-bin matching retains all or none of a clock bin's original members. At zero pairing, the unfiltered common support contains 915,039 second samples; 1,347 have B2 deviations exceeding 100 Hz. The ±100-Hz criterion is a post-inspection gross-excursion screen, with ±10 and ±1000 Hz alternatives retained, rather than a threshold chosen for tidal agreement.

We compare $y = A_T T + Z\eta + \epsilon$ with $y = A'_T T + Z\eta' + \lambda B2 + \epsilon'$. Paired day deletions refit both models and estimate the uncertainty of $A'_T - A_T$ from the same deletions. The monitor's own template projection is also retained. Its frequency deviations are normalized by the nominal carrier 193399200000000 Hz only for a round-trip diagnostic. No division by two or normalization by the 10-MHz RF carrier is used to assert a one-way fractional optical-frequency error.

Overlapping Allan deviations[10] use only contiguous common second samples. A term is formed from the difference of adjacent τ-length averages, and no term crosses a gap. Runs shorter than 4τ are omitted at that averaging time; all contributing term counts are retained. No extrapolation of the monitored summer support to the whole record is assumed. The available timestamps establish labels and pairing conventions, not counter synchronization accuracy. No same-site null comparison is included in this analysis.

Long-period amplitudes and upper bounds

We estimate all five long-period quadrature pairs jointly with one global intercept. Fixed-phase scales are an additional target. Scenarios vary fixed versus jointly fitted short-period terms, one optional global drift, printed versus component-sum corrections, omission of the period with missing May Yb correction, and summer-only support. They introduce no segment-, period- or month-specific intercepts for seasonal inference. A global reference change therefore leaves the target estimates unchanged.

For a pair $\hat{b}$ with covariance C and denominator degrees of freedom ν, a conservative radial bound from the nominal ellipse is

$$U = \|\hat{b}\| + \sqrt{2F_{2,\nu}(1-\alpha)\lambda_{\max}(C)}.$$

We use $\alpha = 0.05$ for an individual bound and 0.01 for the five-target Bonferroni family. With eight period clusters, $\nu = 7$ is a working approximation. The no-drift family upper bounds for Mf, Mm, Msf, Sa and Ssa are respectively $(1.82, 6.06, 3.35, 7.56, 5.05) \times 10^{-16}$. These bounds assume the specified calibration coordinate and do not bound unknown systematic shifts. The summer-only seasonal design is poorly conditioned and yields very large bounds.

The independent solar check averages the degree-two potential over hour angle. With P_2 the second Legendre polynomial, it evaluates $-(1+k_2-h_2)GM_\odot[r_W^2 P_2(\sin\varphi_W) - r_S^2 P_2(\sin\varphi_S)]P_2(\sin\delta)/(r_\odot^3 c^2)$ using DE430 and the recorded geodetic coordinates (Wuhan: 30.529° N, 114.357° E, 50 m; Shanghai: 31.130° N, 121.570° E, 20 m), converted to geocentric radii and latitudes for this calculation. This checks the small Sa/Ssa scale without fitting observations. It does not provide a complete seasonal-loading model. The existing 18-constituent loading series contains none of Mf, Mm, Msf, Sa or Ssa.

Computational reproducibility

The Shanghai–Wuhan observations were acquired independently of the contemporaneous European work, as reported by the experimental team. The new whole-template comparisons, spectral diagnostics and component analyses were developed after that work and the subsequent critique became available. The 1,350-km distance is the fibre route; geographical station separation and clock species are not used to claim an exclusive experimental category.

The analysis retains source-file hashes, selection and exclusion mappings, derived observations, design matrices, covariance outputs, simulation seeds and figure source data. An independent QR/FWL formulation checks the 37 saved mean-model fits and 54 screened monitor comparisons; its maximum prediction difference is 1.63×10^{-10} in 10^{-18} units. Python 3.12.3, NumPy 2.5.4 and SciPy 1.18.1 are used for the principal calculations; astronomical construction uses the existing environment with NumPy 2.5.3 and pyTMD 3.0.9. Matplotlib produces the figures from these outputs. Numerical reproducibility does not constitute independent certification of experimental calibration or statistical coverage.

Data availability

Source manifests, derived time series, exclusion records and figure source data are retained in the versioned analysis package. Access arrangements and any restrictions must be completed by the authors before submission. No public deposit is claimed in this working draft.

Code availability

Analysis, validation and plotting code are retained with their input hashes and execution records. The authors must confirm the access route for editors, referees and readers before submission. No released repository is claimed here.

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